

TEAM 1

Launch

Marble Run Challenge · Tata Advanced Systems Limited

Name	Name
1.	4.
2.	5.
3.	6.

Keep this book on your table. Everything you need is in it.

1. What the room is building

Ten teams. One marble run.

Two balls are launched at the same instant from Team 1. They travel by two separate routes through the room and finish in two buckets at Team 10. The run succeeds when both balls land in their buckets at the same time.

Your team owns one section. It has to take a ball from the team before you and pass it to the team after you. No section is judged on its own — if one fails, the run fails.

2. The two paths

Path A — small ball

Team	Section
1	Launch ← you
2	Raise elevation 1.5 ft above launch
4	Long jump, 1.5 ft
6	Open section
8	Timing
10	Final capture

Path B — large ball

Team	Section
1	Launch ← you
3	Drop elevation 2 ft below launch
5	Short jump, 0.75 ft
7	Open section
9	Timing
10	Final capture

3. Your challenge: Launch

Launch both balls at the same instant, from a single release.

One ball goes to Path A, the other to Path B.

The launch point must be at least 3 ft above the floor.

Both balls must leave with enough speed to complete the rest of the run.

You receive from: —

You hand on to: Team 2 (Path A) and Team 3 (Path B)

You are done when: One action releases both balls together, and each enters the next team's section cleanly at a height you have agreed with them.

Rules that apply to every section

- Once the run starts, nobody touches a ball.
- No motors, no batteries, no compressed air, no external power.

- Anything you build runs on the energy the ball brings, or on energy you store in your section before the run starts.
- Your section stands on its own. Nobody holds it during the run.
- Measurements are checked with a tape measure. Build to the number.

4. What is in your box

Item	Qty
25mm pipe, 1m	20
25mm three-way corner	8
25mm four-way corner	4
25mm tee	8
25mm elbow 90°	5
25mm coupler	3
25mm end cap	4
25mm saddle clamp	6
32mm pipe, 1m	2
32mm elbow 90° (short radius)	4
32mm tee	2
32mm end cap	1
32mm coupler	2
Cardboard blank (sheet)	2
Zip ties (pack of 10)	1
Duct tape	1
Elastic cord set	2
Dowel pack	2
A3 sheets	2
Rubber mallet	1
Ball 12 mm	1
Ball 25 mm	1

Tools

Item	Qty
Tape measure	1
PVC cutter	1
Junior hacksaw	1
Paper knife	1
Permanent pen tip marker	1
Gloves (pairs)	2
Safety glasses	2

Your money

Every team gets the same: **2,000 Kasu**. The materials are not the same.

Coin	Number	Value
500 Kasu	2	1,000
100 Kasu	3	300
50 Kasu	8	400
20 Kasu	10	200
10 Kasu	10	100
Total	33	2,000

No box has everything its team needs. That is deliberate. Somewhere in this room is a team holding what you are short of.

5. What every team has

Your column is highlighted. Put what you don't need in your spares tray so others can see it.

Structural — 25mm

Item	1	2	3	4	5	6	7	8	9	10
25mm pipe, 1m	20	18	24	14	14	6	8	16	14	20
25mm three-way corner	8	6	12	6	6	2	10	6	5	7
25mm four-way corner	4	3	4	.	.	.	3	.	.	3
25mm tee	8	4	8	5	5	2	4	6	5	4
25mm elbow 90°	5	2	6	3	3	1	4	4	3	3
25mm coupler	3	3	8	3	3	1	6	3	2	2
25mm end cap	4	3	8	3	3	1	2	4	3	3
25mm saddle clamp	6	4	5	5	4	2	14	6	5	9

Pathway and consumables

Item	1	2	3	4	5	6	7	8	9	10
32mm pipe, 1m	2	4	4	3	2	1	.	3	3	8
32mm bend 90° (long radius)	.	.	.	.	2	.	.	1	.	1
32mm elbow 90° (short radius)	4	4	2	2	.	1	.	2	3	2
32mm bend 45°	.	.	3	.	.	.	2	.	.	.
32mm Y junction	.	.	.	.	2	.	.	.	.	.
32mm tee	2	1	1	1	.	1	2	1	.	1
32mm end cap	1	1	1	1	1	.	2	1	.	2
32mm coupler	2	2	.	.	.	.	6	.	.	.
32mm × 25mm reducer bush	.	.	.	.	.	.	4	.	.	.
76mm split channel, 1m (SWR Type B)	.	.	3	1	1	.	.	2	1	2
76mm split bend 90° (SWR Type B)	.	.	2	1	.	.	.	1	.	2
Flexible borewell hose, 1m	.	.	.	.	.	.	.	.	5	.
Split foam insulation, 1m	.	.	.	.	.	.	2	.	4	.
Cardboard blank (sheet)	2	4	.	2	8	1	.	3	.	3
Zip ties (pack of 10)	1	4	1	2	4	1	6	3	2	2
Fevicol	.	1	.	.	3	.	.	.	.	1
Scissors	.	1	.	.	3	.	.	.	.	.
Duct tape	1	1	.	.	3	.	.	.	.	1
Elastic cord set	2	.	.	.	.	.	.	.	.	.
Dowel pack	2	.	.	1	.	.	.	.	.	.
Narrow bucket / paper cup	.	.	.	.	.	.	.	.	.	2
Sensor module	.	.	.	.	.	.	.	1	1	.
A3 sheets	2	2	2	2	2	2	2	2	2	2
Permanent pen tip marker	1	1	1	1	1	1	1	1	1	1
Rubber mallet	1	1	1	1	1	1	1	1	1	1
Ball 12 mm	1	1	.	1	.	1	.	1	.	1
Ball 25 mm	1	.	1	.	1	.	1	.	1	1

6. Parts at the depot

Price shown is the starting price. It can go up.

STRUCTURAL

photo 25mm pipe, 1m 60	photo 25mm three-way corner 60	photo 25mm four-way corner 80	photo 25mm tee 40	photo 25mm elbow 90° 40	photo 25mm coupler 20	photo 25mm end cap 20	photo 25mm saddle clamp 40
---	---	--	--	--	--	--	---

PATHWAY CLOSED

photo 32mm pipe, 1m 200	photo 32mm coupler 60	photo 32mm elbow 90° (short radius) 100	photo 32mm bend 90° (long radius) 300	photo 32mm bend 45° 180	photo 32mm tee 140	photo 32mm Y junction 220	photo 32mm end cap 40	photo 32mm x 25mm reducer bush 120
--	--	--	--	--	---	--	--	---

PATHWAY OPEN

photo 76mm split channel, 1m (SWR Type R) 200	photo 76mm split bend 90° (SWR Type B) 240
--	---

FLEX & FRICTION

photo Flexible borewell hose, 1m 300	photo Split foam insulation, 1m 120
---	--

CONSUMABLES

photo Cardboard blank (sheet) 120	photo Zip ties (pack of 10) 80	photo Fevicol 200	photo Scissors 240	photo Duct tape 160	photo Elastic cord set 140	photo Dowel pack 100	photo Permanent pen tip marker 60	photo Rubber mallet 300
photo Funnel 200	photo Narrow bucket / paper cup 100	photo A3 sheets 20						

SPECIAL

photo Sensor module 200
--

7. The depot

- The depot sells at the price on the board. The board is the truth, not this book.
- It buys back at the **starting price**, whatever the board says at the time.
- Unused and in original condition: full price back. Cut or modified but usable: half. Cut about and unusable: nothing.
- **Only your nominated Trader may approach the counter.**

The cutting bay

- Bring material and it will be cut for you on power tools. No charge.
- Up to five cuts per visit, then back of the queue.
- **Nobody from your team touches the power tools.** Stand outside the marked line.
- You have a cutter and hacksaw in your box. The bay is faster and cleaner. Whether the queue is worth it is your call.

8. Trading with other teams

No rules beyond common sense. Sell, buy, swap, lend, pool money, split a purchase, trade a favour or a pair of hands.

You cannot trade your safety gear. Everything else is yours to deal with.

Write down what you agree. There is a page at the back.

9. Things will change

Prices at the depot are not fixed. Events during the game can move them, sometimes a long way. When something changes we will announce it and the board will be updated. Keep an eye on it.

Prices are not the only thing that can change. Treat any announcement as part of the game.

10. Working with the material

Safety. Gloves and glasses on whenever anyone is cutting. Cut away from your body, and keep cutting to one end of your table.

Cutting. Measure twice. Cut square, and clean the burr off the inside and outside of every cut edge before you assemble anything.

Assembly. Dry-fit before you commit to glue. Glue is permanent and glued parts have no resale value. Zip ties and tape hold more than you think.

Testing. Roll a ball through each piece as you finish it. Do not wait until the end.

Clearance. Every opening, catcher and channel must let the gauge through. Check it physically, not by eye. You may be asked to run a larger ball later.

11. Trade log

[illegible]

12. Notes

[illegible]